

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 5

(1) Let R be a commutative ring and let $I \leq R$ be an ideal.

(a) Show that the assignment $J \mapsto J/I$ is a bijection

$$\{\text{Ideals of } R \text{ containing } I\} \leftrightarrow \{\text{Ideals of } R/I\}.$$

(b) Show that, given ideals $I \leq J \leq R$, there is an isomorphism of rings

$$R/J \xrightarrow{\cong} (R/I)/(J/I).$$

Solution. Note that by the Third isomorphism theorem, shown last semester, we know that there is a bijection $J \mapsto J/I$

$$\{(\text{Additive}) \text{ subgroups of } R \text{ containing } I\} \leftrightarrow \{\text{Subgroups of } R/I\}$$

We want to show that the property of being an *ideal* is matched up on both sides. For this, note that we have the following diagram

$$\begin{array}{ccc} R & & \\ \downarrow & \searrow I \leq J & \\ R/I & \longrightarrow & R/J \end{array}$$

Applying the factoring triangle to the bottom arrow again, and noting that $\ker(R/I \rightarrow R/J) = J/I$ gives us

$$\begin{array}{ccc} R/I & & \\ \downarrow & \searrow & \\ (R/I)/(J/I) & \xrightarrow{\cong} & R/J \end{array}$$

Now, we know that R/I is a ring, and the bottom arrow is an isomorphism. Therefore, the domain is a ring (equivalently, $J/I \leq R/I$ is an ideal) if and only if the codomain R/J is a ring (equivalently, $J \leq R$ is an ideal). In this case, the bottom arrow is an isomorphism of rings. Therefore, we have simultaneously proved (a) and (b).

Definition 1. A proper ideal $I \leq R$ in a commutative ring is **prime** if for all $a, b \in R$, $ab \in I$ implies that either $a \in I$ or $b \in I$. It is **maximal** if there is no larger proper ideal of R containing I .

(2) Show that the following are equivalent for a proper ideal $I \leq R$:

- (a) I is prime.
- (b) R/I is an integral domain.

Solution. Saying that R/I is an integral domain is the same as saying: $(a + I)(b + I) = ab + I = I$ implies either $a + I = I$ or $b + I = I$. This translates exactly to: $ab \in I$ implies either $a \in I$ or $b \in I$.

(3) Show that the following are equivalent for a proper ideal $I \leq R$:

- (a) I is maximal.
- (b) R/I is a field.

Solution. By problem 1(a), saying that I is maximal is the same as saying that the only ideals in R/I are (0) and R/I . So if we set $S = R/I$, then the problem translates to: S is a field if and only if its only ideals are (0) and S .

We already know that the ‘only if’ direction: A field S has no proper, non-trivial ideals (why?).

For the ‘if’ direction: We need to show that every non-zero element of S is invertible. Pick $a \in S \setminus \{0\}$. Now, consider the principal ideal $(a) \leq S$: This is non-zero, and so by our assumption has to be all of S . Therefore, we have $1 \in (a)$, which means $1 = as$ for some $s \in S$. This of course means that 1 is invertible.

A finitely generated k -vector space V will be called a **finite dimensional k -vector space**, and its **k -dimension** is the unique integer d such that V is isomorphic to k^d . We will denote this integer by $\dim_k V$.

- (4) Suppose that $T : V \rightarrow W$ is a linear transformation of finite dimensional k -vector spaces. Show that

$$\dim_k(\ker T) + \dim_k(\operatorname{im} T) = \dim_k V.$$

*This is usually referred to as **rank-nullity**.*

Solution. The first isomorphism theorem gives us an isomorphism $V/\ker T \xrightarrow{\cong} \operatorname{im} T$. This tells us

$$\dim_k(V/\ker T) = \dim_k(\operatorname{im} T).$$

On the other hand, if we fix an isomorphism $V \xrightarrow{\cong} k^n$ where $n = \dim_k V$, then $\ker T$ is mapped into a subspace $U \leq k^n$ with $\dim_k U = \dim_k(\ker T)$. By Proposition 2 of Lecture 15, we find

$$\dim_k(V/\ker T) = \dim_k(k^n/U) = n - \dim_k U = \dim_k V - \dim_k(\ker T).$$

Equating this with the dimension of the image as above gives us rank-nullity.

Remark 1. The main point of this is the equality

$$\dim_k(V/W) = \dim_k V - \dim_k W$$

for a subspace $W \leq V$ of a finite dimensional k -vector space V .

- (5) Find all polynomials of degree 6 over k that vanish at 0, 1 and 2.

Hint: Let $P_6(k)$ be the space of polynomials of degree at most 6. Then you’re computing the kernel of a particular linear transformation from P_6 to k^3 . Note that when k has characteristic 2, you might have a different answer than in general.

Solution. Consider the function

$$T : P_6(k) \xrightarrow{p(x) \mapsto (p(0), p(1), p(2))} k^3.$$

It is not hard to see that this is a linear transformation. If we use the basis given by $(1, x, \dots, x^6)$ for $P_6(k)$ and the standard basis for k^3 , then you can check that the linear transformation corresponds to the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 & 32 & 64 \end{bmatrix},$$

To compute the kernel, we can use the reduced row echelon form from linear algebra. What is the r.r.e.f of this matrix? Well, it depends on whether $2 = 0 \in k$! If $2 \in k^\times$, then the r.r.e.f is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 & -6 & -14 & -30 \\ 0 & 0 & 1 & 3 & 7 & 15 & 31 \end{bmatrix}.$$

The kernel of this matrix is

$$\left\{ \begin{bmatrix} 0 \\ 2c_1 + 6c_2 + 14c_3 + 30c_4 \\ -3c_1 - 7c_2 - 15c_3 - 31c_4 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} : c_1, c_2, c_3, c_4 \in k \right\}.$$

If is not clear how I got this kernel, ask me!

Note that $P_6(k)$ is a 7-dimensional space, and we are imposing three conditions on it, so you should expect to get a four-dimensional space after that.

This translates to saying that the polynomials we are looking for are those of the form

$$\{(2c_1 + 6c_2 + 14c_3 + 30c_4)x - (-3c_1 - 7c_2 - 15c_3 - 31c_4)x^2 + c_1x^3 + c_2x^4 + c_3x^5 + c_4x^6 : c_1, c_2, c_3 \in k, c_4 \in k^\times\}.$$

Here, the condition $c_4 \in k^\times$ ensures that we are only looking at polynomials of degree 6 and not any lower.

If $2 = 0 \in k$, then the r.r.e.f is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

and we instead get the polynomials of the form

$$\{-(c_1 + c_2 + c_3 + c_4 + c_5)x + c_1x^2 + c_2x^3 + c_3x^4 + c_4x^5 + c_5x^6 : c_1, c_2, c_3, c_4 \in k, c_5 \in k^\times\}.$$

What happened here is that what were three independent conditions in general now become two, because $p(0) = 0$ and $p(2) = 0$ are saying the same thing when $2 = 0$.

Remark 2. Alternatively, we can just note that the conditions are telling us that we are looking for polynomials with 0, 1, 2 as zeros, which means precisely that they are multiples of $x(x-1)(x-2)$ (when the characteristic is not 2) and of $x(x-1)$ (when the characteristic is 2).

Suppose that $k \leq L$ is an extension of fields (that is, one field contained in another as a sub-field). We will say that L is a **finite extension of k** if L is a finite dimensional k -vector space. In this case, we will set $[L : k] = \dim_k(L)$, and call it the **degree** of L over k .

- (6) Suppose that $k \leq L$ is a finite field extension of degree n . Show that every finite dimensional L -vector space V is finite dimensional as a k -vector space, and that

$$\dim_k V = n \dim_L V.$$

Solution. Since $[L : k] = n$, we have an isomorphism

$$k^n \xrightarrow{\sim} L$$

of k -vector spaces. For any integer d , this gives us an isomorphism

$$k^{nd} \simeq (k^n)^d \xrightarrow{\sim} L^d$$

of k -vector spaces.

If $\dim_L V = d$, then we have an isomorphism

$$L^d \xrightarrow{\sim} V$$

of L -vector spaces. Any such isomorphism is automatically an isomorphism of k -vector spaces (note that V obtains the structure of a k -vector space, because being able to scale by any element of L implies in particular that you can scale by any element of k). Combining this with the previous isomorphism gives an isomorphism

$$k^{nd} \xrightarrow{\sim} L^d \xrightarrow{\sim} V$$

of k -vector spaces, showing $\dim_k V = nd = n \dim_L V$.

- (7) Use the previous problem to prove the following facts, which we saw in a different way earlier this term:

- (a) If k is a finite field of characteristic p , then $|k| = p^r$ for some prime p and $r \in \mathbb{Z}_{\geq 1}$.

Solution. By assumption, k contains $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, and is thus a field extension of $\mathbb{F}_p$. Since k has only finitely many elements, it must be generated by a finite subset, and so is a finite dimensional $\mathbb{F}_p$ -vector space, isomorphic to $\mathbb{F}_p^r$ for some r , where $r = [k : \mathbb{F}_p]$ is the degree of the extension. But $\mathbb{F}_p^r$ has p^r elements (why?), and so k must also have p^r elements.

- (b) If $k \leq L$ is an extension of finite fields with $|k| = p^r$ and $|L| = q^s$ then $p = q$ and $r \mid s$.

Solution. Note that k contains $\mathbb{F}_p$ and L contains $\mathbb{F}_q$, for *unique* primes p and q . Since L also contains k (and hence $\mathbb{F}_p$), this means $p = q$.

Furthermore, L is a finite extension of k and so is isomorphic to k^t for some t . This means that it has $|k|^t = p^{rt}$ elements, which shows that $s = rt$ is a multiple of r .

An n -tuple $(v_1, \dots, v_n)$ in a k -vector space V is **k -linearly independent** (or simply **linearly independent**) if the only solution in $a_1, \dots, a_n \in k$ for the equation $a_1 v_1 + \dots + a_n v_n = 0$ is $a_1 = a_2 = \dots = a_n = 0$. The n -tuple is a **basis** if $\{v_1, \dots, v_n\}$ generates V as a k -vector space.

- (8) Let $T : k^n \rightarrow V$ be the unique linear transformation defined by $T(e_i) = v_i$. Show:

- (a) T is injective if and only if $(v_1, \dots, v_n)$ is linearly independent.

Solution. This is more or less definitional: The vector $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ is in $\ker T$ precisely when $a_1 v_1 + \dots + a_n v_n = 0$, so saying that $\ker T = \{0\}$ is the same as saying that the vectors are linearly independent.

- (b) T is an isomorphism if and only if $(v_1, \dots, v_n)$ is a basis.

Solution. More generally, T is surjective if and only if $\{v_1, \dots, v_n\}$ generates V as a k -subspace: Indeed, its image is exactly the subspace of V generated by the vectors.

- (c) If $\dim_k V = n$, then the following are all equivalent:

- (i) $(v_1, \dots, v_n)$ is a basis.
- (ii) $(v_1, \dots, v_n)$ is linearly independent.
- (iii) $\langle \{v_1, \dots, v_n\} \rangle_k = V$.

Solution. In this case, $T : k^n \rightarrow V$ is a linear transformation between vector spaces of the same dimension. Applying rank-nullity says

$$\dim \ker T + \dim \operatorname{im} T = n$$

Therefore, $\dim \operatorname{im} T = n = \dim_k V$ if and only if $\dim \ker T = 0$. In other words, T is injective if and only if it is surjective, and so injectivity and surjectivity are both equivalent to T being an *isomorphism*. Combining this with parts (a) and (b) tells us $(1) \Rightarrow (2) \Leftrightarrow (3)$ and $(2) \Rightarrow (1)$.

- (9) Suppose that $k \leq L$ is a field extension and that W is an L -vector space. Is it true that every k -linearly independent subset of W is also L -linearly independent? If true, provide a proof; if false, provide a counter-example.

Solution. This is *false*. Take $k = \mathbb{R}$, $L = \mathbb{C}$ and $W = \mathbb{C}$ to be the 1-dimensional space over $\mathbb{C}$. The vectors $1, i \in W$ are $\mathbb{R}$ -linearly independent (they are not real multiples of each other), but are not $\mathbb{C}$ -linearly independent.

Suppose that V is a finite dimensional k -vector space and $\mathcal{B} = (v_1, \dots, v_d)$ is a basis for V . Let $\psi_{\mathcal{B}} : k^d \xrightarrow{\cong} V$ be the isomorphism associated with this basis. The $\mathcal{B}$ -matrix for a linear transformation $T : V \rightarrow V$, denoted $[T]_{\mathcal{B}}$ is the $d \times d$ -matrix that represents the composition

$$k^d \xrightarrow[\cong]{\psi_{\mathcal{B}}} V \xrightarrow{T} V \xrightarrow[\cong]{\psi_{\mathcal{B}}^{-1}} k^d$$

- (10) Let $f(x) \in k[x]$ be a non-constant monic polynomial of degree d . Let $\mathcal{B}$ be the basis for $k[x]/(f(x))$ given by the cosets $(1+(f(x)), x+(f(x)), \dots, x^{d-1}+(f(x)))$. Compute the $\mathcal{B}$ -matrix for the multiplication-by- x linear transformation

$$T : k[x]/(f(x)) \xrightarrow{r(x)+(f(x)) \mapsto xr(x)+(f(x))} k[x]/(f(x)).$$

Solution. Concretely, the matrix $[T]_{\mathcal{B}}$ has the following property: If $v_i \in \mathcal{B}$ and $T(v_j) = \sum_i a_{i,j} v_i$, then $[T]_{\mathcal{B}} = [a_{i,j}]_{1 \leq i, j \leq n}$ is the matrix whose (i, j) -th entry (i -th row and j -th column) is $a_{i,j}$. Given this, in our particular case, let's set $v_j = x^{j-1} + (f(x))$ with $j = 1, \dots, n$. Suppose that

$$f(x) = a_d x^d + a_{d-1} x^{d-1} + \dots + a_0 \in k[x]$$

with $a_d \neq 0$ and $d \geq 1$. Note that we have

$$x^d + (f(x)) = x^d - a_d^{-1} f(x) + (f(x)) = -a_d^{-1} a_0 - \dots - a_d^{-1} a_{d-1} x^{d-1} + (f(x)) = -a_d^{-1} a_0 \cdot v_1 - \dots - a_d^{-1} a_{d-1} v_d$$

Then we see

$$T(v_j) = x \cdot (x^{j-1} + (f(x))) = x^j + f(x) = \begin{cases} v_{j+1} & \text{if } j \leq d-1 \\ -a_d^{-1} a_0 \cdot v_1 - \dots - a_d^{-1} a_{d-1} v_d & \text{if } j = d. \end{cases}$$

Therefore $[T]_{\mathcal{B}}$ looks like:

$$[T]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_d^{-1} a_0 \\ 1 & 0 & \dots & 0 & -a_d^{-1} a_1 \\ 0 & 1 & \dots & 0 & -a_d^{-1} a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_d^{-1} a_{d-1} \end{bmatrix}$$